The Dark Triad as a Multi-Layered Structure: A Triangulation of Classical Psychometrics and Explainable Machine Learning for Testing the Distinctiveness and Convergence of Machiavellianism, Psychopathy, and Narcissism

Sabihe Khorshidi[1] and Kian Mansouri Jamshidi[2]

[1]Psychology Specialist, Project Lead; Email: sabihekhorshidy@gmail.com
[2]Head of AI Division; Email: sirkianmj@gmail.com

Author Note
This study is a secondary analysis of de-identified data from the Czech validation study of the Dark Triad Dirty Dozen (DTDD), originally collected by the Olomouc University Social Health Institute (OUSHI), Palacký University in Olomouc, Czech Republic. The original data collection was conducted with adult participants who provided informed consent and received ethical approval from the Ethics Committee of the Faculty of Theology of Palacký University in Olomouc (Approval No. 2020/4). The present authors did not recruit participants or collect the original data. The present manuscript reports analyses conducted on the previously collected and publicly available dataset. No new participant recruitment or primary human-subject data collection was undertaken for the present secondary analysis.

Abstract

Since its introduction, the Dark Triad—Machiavellianism, psychopathy, and narcissism—has raised the question of whether these traits are distinct or manifestations of a shared Dark Core. Prior work has relied on linear tools such as CFA/OLS, which cannot detect nonlinearities. Using existing Dirty Dozen data from community ($N = 5{,}902$), student ($N = 2{,}071$), and representative ($N = 1{,}492$) samples, we applied the Dark Triad Triangulation (DT^3) framework across five layers: network psychometrics, supervised machine learning (XGBoost, multi-task neural network), counterfactual tipping points, semantic embeddings, and robustness testing. CFA supported three factors (CFI$= 0.941$, RMSEA$= 0.077$), but latent factor correlations were high ($r = 0.62, 0.61$). The 12-item network separated into three communities; when embedded in the full 88-item personality space, all Dark Triad items merged into one Dark Core. Supervised predictors diverged (agreeableness for Machiavellianism, empathy for psychopathy, age for narcissism), CKA indicated representational distinctiveness, and semantic clustering partially separated traits (ARI$= 0.505$), though narcissism overlapped with Machiavellianism. Out-of-sample R^2 (0.13–0.23) limits inferences. Overall, findings support a middle ground: three distinguishable dimensions nested within a shared higher-order structure.

Keywords: Dark Triad of personality; Dark Core of Personality; explainable machine learning; SHAP; network psychometrics; Machiavellianism; psychopathy; narcissism; out-of-sample validation

1 Introduction

Since Paulhus & Williams (2002) first grouped together the three subclinical traits of Machiavellianism, psychopathy, and narcissism under the label "Dark Triad," this framework has become one of the most widely used research constructs in personality, organizational, and social psychology. Much of its appeal derives from a foundational hypothesis: although the three traits correlate with one another, each represents a relatively distinct behavioral, motivational, and interpersonal pattern. Brief instruments such as the 12-item Dirty Dozen (DTDD; Jonason & Webster, 2010) and the Short Dark Triad (SD3; Jones & Paulhus, 2014) enabled this research to be conducted at scale in non-clinical populations.

From the outset, a serious controversy emerged regarding the true independence of these traits. A meta-analysis by Vize et al. (2018) showed that although the nomological networks of the three traits differ, these differences are often small and situated at the margins of a larger shared core; the Machiavellianism–psychopathy overlap in particular is so pronounced that Miller et al. (2017) explicitly asked whether it constitutes "a distinction without a difference." In response, Moshagen et al. (2018) proposed the "Dark Factor of Personality" (D): a basic tendency to maximize one's own utility at others' expense, accompanied by corresponding justifying beliefs. From this perspective, Machiavellianism, psychopathy, narcissism, and related dark traits are surface manifestations of a single underlying disposition rather than fully independent constructs.

Empirical work on this controversy has relied almost entirely on classical linear methods — CFA and OLS regression. These methods have two important limitations. First, both assume linear, additive relationships and are thus ill-suited to detecting thresholds, higher-order interactions, or nonlinear functional forms. Second, latent factor correlations — typically used to adjudicate distinctiveness — are a summary, unidimensional index that cannot reveal whether the predictive pattern, internal representation, or semantic struc-

ture of the three traits genuinely differ. Recent developments in explainable AI and network psychometrics provide complementary, nonlinear vantage points.

To fill this gap, the present study introduces the "Dark Triad Triangulation" (DT^3) framework. Its logic is straightforward: if the three traits are genuinely distinct, this distinctiveness should be convergently observable across several independent types of evidence, not merely within a single statistical test. Five layers of computational evidence are applied to existing DTDD data: (1) unsupervised network psychometrics of item partial correlations; (2) supervised divergence via XGBoost and neural representational similarity (CKA); (3) counterfactual tipping-point analysis; (4) semantic embeddings of item text via pretrained language models; and (5) robustness testing across algorithmic architectures and samples. These layers, alongside baseline psychometrics (reliability, temporal stability, CFA, regression), form a complementary evidentiary base for testing "three independent traits" versus "one shared dark core."

This study constitutes a secondary, exploratory, non-preregistered analysis of existing data. Its aim is not to test a single predetermined hypothesis but to demonstrate the integrative capabilities of psychometrics and explainable machine learning for classic questions of personality structure — a template applicable to other domains of psychometrics.

1.1 Literature Review and Background

Paulhus & Williams (2002) proposed the term "Dark Triad" for the co-occurrence of three subclinical yet socially aversive traits. Machiavellianism reflects a cynical view of human nature and instrumental exploitation of others; narcissism reflects superiority, entitlement, and need for admiration; and psychopathy reflects low empathy/guilt, affective callousness, and disregard for consequences. Despite differing content, the three traits show meaningful overlap.

Multiple meta-analyses document this overlap. Muris et al. (2017) showed that the three traits share substantial intercorrelations and are primarily associated with low agreeableness (FFM) and low honesty-humility (HEXACO), and are more pronounced in men; Muris et al. (2017) also found that, unlike the other two, narcissism showed positive associations with extraversion, conscientiousness, and openness. O'Boyle et al. (2015), based on more than 300 independent samples, found that although the Five-Factor Model accounts for substantial Dark Triad variance, each trait contributes unique information beyond the Big Five. Vize et al. (2018) showed that nomological-network differences among the traits are meaningful but sit at the margins of a shared core, and Miller et al. (2017) explicitly questioned whether Machiavellianism and psychopathy are distinguishable.

The growth of this literature owes much to brief instruments: the Dirty Dozen (Jonason & Webster, 2010) and the SD3 (Jones & Paulhus, 2014). More recent evidence indicates greater structural complexity: Knitter et al. (2025), using meta-analytical structural equation modeling, found that conventional correlated three-factor and orthogonal bi-factor models fail to fully capture irregular item correlations; for the Dirty Dozen, a bifactor-(S·I-1) model fit best, with Machiavellianism–psychopathy overlap (especially vindictiveness items) as the primary source of ambiguity.

In response to this overlap, Moshagen et al. (2018) proposed the Dark Core theory: dark traits are surface manifestations of a shared tendency to maximize utility at others' expense, plus justifying beliefs. This predicts a hierarchical structure combining a shared core with trait-specific dimensions.

Network psychometrics offers a complementary lens. Papageorgiou et al. (2019) found that narcissism, unlike the other traits, occupies a bridging position between prosocial and antisocial personality poles. Wehner et al. (2021) found that interpersonal manipulation (a psychopathy facet) is the most central network node, and that adding Big Five facets only marginally alters overall network structure, though dark traits still cluster by theoretical domain.

Empathy correlates with the Dark Triad non-uniformly. Shukla & Upadhyay (2025), synthesizing 14 studies ($N = 5{,}328$), found narcissism negatively related to affective but not cognitive empathy, Machiavellianism negatively related to both, and psychopathy showing the strongest negative relation to affective empathy.

Machine learning offers a complementary lens to CFA/OLS. Leberecht et al. (2026) compared ML methods for predicting Dark Triad traits from social-media text, showing that predictive accuracy and bias vary by algorithm — though this capacity has rarely been paired with explainability tools such as SHAP for direct trait comparison.

Overall, meta-analytic/psychometric evidence (Muris et al., 2017; O'Boyle et al., 2015; Knitter et al., 2025) and Dark Factor theory (Moshagen et al., 2018) support a substantial shared core, while network, nomological, and empathy evidence (Papageorgiou et al., 2019; Wehner et al., 2021; Shukla & Upadhyay, 2025) indicates retained distinctiveness — motivating a multi-method approach.

1.2 Statement of the Problem and Research Gap

Despite two decades of research, it remains unclear to what extent the three traits are distinct versus manifestations of a shared core. If genuinely distinct, distinctiveness should appear not only in factor structure but in relationships with other dimensions, predictive patterns, interactions, and computational/semantic representations.

Existing evidence relies almost entirely on CFA and linear regression, which assume linear/additive relationships and reduce distinctiveness to a single summary correlation. Seven specific gaps motivate this study: (1) uncertainty whether observed distinctiveness reflects true structure or instrument artifacts (Knitter et al., 2025); (2) unclear boundary stability when the Dark Triad is embedded within a broader personality network (Papageorgiou et al., 2019; Wehner et al., 2021); (3) reliance on pairwise correlations rather than multivariate functional comparison (O'Boyle et al., 2015; Muris et al., 2017; Shukla & Upadhyay, 2025); (4) neglect of non-

linear/interactive relationships (Leberecht et al., 2026); (5) paucity of explainable-AI evidence on feature-importance structure; (6) absence of cross-method triangulation; and (7) untested robustness across algorithms and samples.

This motivates a testable middle-ground hypothesis: Machiavellianism, psychopathy, and narcissism share a meaningful common component (consistent with the Dark Core theory; Moshagen et al., 2018) while retaining distinctive patterns across psychometric structure, relational networks, predictive drivers, feature importance, and computational representations.

1.3 Aims and Research Questions

This study addresses three questions:

1. When examined with supervised and unsupervised methods beyond linear factor analysis, do the three traits exhibit distinctive predictive patterns, internal representations, and semantic structures?
2. How does this distinctiveness change when the variable space expands from the 12 DTDD items to the full personality space (Big Five, empathy)?
3. Which trait, if any, shows the greatest functional overlap with another, and with which personality constructs is this overlap associated?

The primary innovation is the integration of five independent computational layers within a single, reproducible DT^3 framework, enabling direct comparison across methods without sample or tool confounds.

2 Methods

2.1 Participants and Samples

Data were drawn from an existing secondary dataset collected for a psychometric validation study of the DTDD (Živná et al., 2025) and made available on OSF. Raw files were reprocessed to ensure feature alignment across analysis layers; reported sample sizes may therefore differ slightly from the original validation study due to inclusion/exclusion criteria.

Three independent samples were obtained after preprocessing: a community sample ($N = 5{,}902$), a student sample ($N = 2{,}071$), and a national representative sample ($N = 1{,}492$; total $N = 9{,}465$). Case-wise exclusion for missing auxiliary data (particularly the Rosenberg Self-Esteem Scale, RSES, essentially absent in the representative sample; exclusion threshold $> 90\%$ missing) reduced analytic Ns per analysis (e.g., Layer 5 cross-validation for the community sample: $N = 5{,}215$). An independent subsample of 61 matched participants (from 3,913 test–retest records) was used for ICC estimation.

The community sample served as the training set and the representative sample as a fully held-out test set for all supervised components (Layers 2, 3, 5); the representative sample was untouched during fitting, hyperparameter search, or feature selection, and used only once for final scoring. The student sample lacked auxiliary variables (Big Five, empathy, demographics) required for feature alignment and was excluded from predictive-modeling layers, retained only for analyses not requiring the full auxiliary set (e.g., reliability).

2.2 Instruments

The Dark Triad was measured with the Dirty Dozen (DTDD; Jonason & Webster, 2010), 12 items on a 7-point Likert scale, four items per trait: Machiavellianism (manipulativeness, deceitfulness, flattery, instrumental exploitation), psychopathy (lack of remorse, moral indifference, callousness, cynicism), and narcissism (need for admiration, attention-seeking, status-seeking, entitlement). The Big Five was assessed with the BFI, empathy with the Toronto Empathy Questionnaire (TEQ), and self-esteem with the RSES (excluded from final integrative analyses due to missingness).

2.3 Data Preprocessing

Steps included: (a) restricting age to 18–100 years; (b) removing flagged "speeders"/low-quality responders; (c) using a minimum item count for composite scores rather than zero-imputation; and (d) requiring $\approx 80\%$ minimum item completion for BFI/TEQ/RSES composites.

2.4 Baseline Psychometrics

Internal consistency used Cronbach's α and McDonald's ω; temporal stability used ICC(2,1) on the test–retest subsample; factor structure used CFA comparing one-, two-, and three-factor models, reporting χ^2, df, CFI, TLI, and RMSEA. Nomological networks were estimated via standardized OLS regression (predictors: Big Five, empathy, self-esteem). Analyses used Python (`semopy` for CFA).

2.5 Computational Methodology: The DT^3 Triangulation Framework

2.5.1 Out-of-Sample Evaluation Strategy

All supervised components (Layers 2, 3, 5) were trained exclusively on the community sample and evaluated once on the held-out representative sample; architectures, hyperparameters, and preprocessing were finalized via internal cross-validation within the community sample only. Analyses combined Python 3 (`xgboost`, `shap`, `scikit-learn`, `sentence-transformers`, `gplearn`) and R (`qgraph`/`EBICglasso`, `lavaan`).

2.5.2 Layer 1: Unsupervised Structural Networks

Structural relationships were modeled via Graphical LASSO regularized with EBIC (EBICglasso; Epskamp & Fried, 2018), with Louvain community detection (Blondel et al., 2008). Two specifications were estimated: a DTDD-only network (12 items) and a full-item network (88 items: Dark Triad plus Big Five and empathy). For the full-item network,

the correlation matrix was projected onto the nearest positive-definite matrix ($\alpha = 0.0346$, max. 1000 iterations) prior to estimation, an explicit limitation discussed in Section 4.5.

2.5.3 Layer 2: Supervised Divergence and Representational Similarity

Trait scores were modeled with Gradient-Boosted Decision Trees (XGBoost; Chen & Guestrin, 2016), chosen for detecting nonlinear thresholds and interactions without OLS assumptions. Separate regressors per trait used an identical predictor set (age, Big Five, TEQ). A Shared-Trunk Multi-Task Neural Network (input $\rightarrow$ 128-d projection $\rightarrow$ two residual blocks $\rightarrow$ dropout $\rightarrow$ three output heads; Adam, lr $= 0.001$, weight decay $1e{-}4$, batch size 128, 300 epochs, `ReduceLROnPlateau`) was trained in parallel. Representational geometry across trait-specific pathways was compared via Centered Kernel Alignment (CKA; Kornblith et al., 2019), benchmarked against a label-permutation null (network retrained per permutation). Symbolic regression via evolutionary search (Fisher et al., 2019) additionally identified trait-specific functional forms without a priori shape constraints.

2.5.4 Layer 3: Counterfactual Tipping-Point Analysis

A deterministic grid-search perturbation method (Wachter et al., 2017) quantified the minimal single-feature change required to shift an above-median case below the median, holding all other features constant. Age was held immutable and excluded from the search space. The outcome metric was the percentage of above-median cases for which perturbing the primary feature sufficed to cross below the median.

2.5.5 Layer 4: Semantic Embeddings

Item text was encoded with `all-MiniLM-L6-v2` (SentenceTransformer family; Reimers & Gurevych, 2019), followed by hierarchical agglomerative clustering (cosine distance, average linkage; $k = 3$ fixed a priori). Cluster assignments were compared to a priori trait labels via the Adjusted Rand Index (ARI), benchmarked against a 1,000-permutation null.

2.5.6 Layer 5: Feature Importance and Robustness Testing

Local/global feature importance used SHAP (TreeExplainer; Lundberg & Lee, 2017). A SHAP Divergence Index (SDI) — a project-specific metric, mean pairwise cosine distance among mean absolute SHAP vectors — was benchmarked against a null built from 500 refits on randomly shuffled/recombined "pseudo-traits" derived from the same 12 items (limitation: not fully independent of the measurement system; Section 4.5). Individual heterogeneity was assessed via Gaussian Mixture Models on local SHAP vectors (BIC-selected components). Architecture robustness was assessed via a Rashomon Set analysis comparing out-of-sample R^2 across XGBoost, Random Forest, and Elastic-Net on an identical split.

2.6 Ethical Considerations, Reproducibility, and Data/Code Availability

This study is a secondary analysis of existing, de-identified data from the Czech validation study of the Dark Triad Dirty Dozen (DTDD). The original human-subject research was conducted by the Olomouc University Social Health Institute (OUSHI), Palacký University in Olomouc, Czech Republic. Adult participants provided informed consent, and the original study received ethical approval from the Ethics Committee of the Faculty of Theology of Palacký University in Olomouc (Approval No. 2020/4). The present authors were not involved in participant recruitment or the original data collection and did not conduct any new human-subject research. The present manuscript reports only secondary analyses of the previously collected, de-identified dataset. No new participant recruitment, intervention, or primary human-subject data collection was undertaken for this study.

Complete preprocessing and analysis code, validation materials, execution logs, and results tables are maintained in a public repository, with software versions documented through environment files (e.g., `renv.lock`). As the present analysis is exploratory and non-preregistered, the findings are hypothesis-generating.

3 Results

3.1 Baseline Psychometrics

Table 1: Internal consistency (α/ω) by trait and sample

Trait	Community	Student	Representative
Machiavellianism	acceptable–good	acceptable–good	acceptable–good
Narcissism	acceptable–good	acceptable–good	acceptable–good
Psychopathy	lower	$\omega = 0.638$	lower

Note. Exact per-cell α/ω values beyond those explicitly reported in text (psychopathy $\omega = 0.638$, student sample) are archived in the project repository; qualitative descriptors reflect the pattern reported in the source analysis.

Table 2: Comparison of fit for one-, two-, and three-factor models (community sample)

Model	CFI	TLI	RMSEA
One-factor	poor fit	poor fit	poor fit
Two-factor	n.r.	n.r.	n.r.
Three-factor	0.941	> 0.90	0.077

Note. n.r. = not reported in source text; full fit indices archived in the project repository.

Table 3: Latent factor correlations in the three-factor model

	Psychopathy	Narcissism
Machiavellianism	0.62	0.61
Psychopathy	—	lowest of the three

Table 4: Standardized OLS regression summary (predictors: Big Five, empathy, self-esteem)

Trait	Adjusted R^2	Dominant predictor(s)
Machiavellianism	n.r.	low agreeableness
Psychopathy	0.278	low agreeableness, low empathy
Narcissism	0.106	positive extraversion

Psychopathy showed consistently lower reliability than Machiavellianism and narcissism, reaching an unfavorable level in the student sample ($\omega = 0.638$). Temporal stability (61 test–retest pairs) replicated this pattern: ICC(2,1) $= 0.826$ (95% CI $[0.726, 0.892]$) for Machiavellianism, 0.836 ($[0.741, 0.899]$) for narcissism, and only 0.637 ($[0.464, 0.767]$) for psychopathy (all $p < .001$). This convergence of lower internal consistency and lower temporal stability is discussed in Section 4.5 as a psychometric limitation of the DTDD psychopathy subscale.

The three-factor model clearly outperformed the competing models and exceeded conventional thresholds (CFI/TLI > 0.90; RMSEA < 0.08), supporting three distinct factors over a single unitary factor.

Machiavellianism shows relatively high correlations with both other traits, while psychopathy and narcissism show the least latent overlap — a bridging pattern for Machiavellianism that recurs in the computational layers (particularly Layers 1 and 4).

Psychopathy showed the best predictability (adjusted $R^2 = 0.278$) and the strongest dependence on low empathy; narcissism, uniquely, showed a positive relationship with extraversion and the lowest explained variance (adjusted $R^2 = 0.106$). Low agreeableness was the shared dominant predictor for Machiavellianism and psychopathy.

3.2 Layer 1: Unsupervised Structural Networks

The EBICglasso network on the 12 DTDD indicators produced three Louvain communities corresponding exactly to the three theoretical traits, confirming the three-factor structure from an independent network-psychometric perspective. When re-estimated in the full 88-item space (following positive-definite projection, $\alpha = 0.0346$), all Dark Triad items merged into a single Louvain community — termed the "Dark Core Collapse." The boundaries of the three traits are observable only when artificially isolated from broader personality context; once embedded in a wider nomological space, these boundaries dissolve into a unidimensional, shared structure. This result carries the matrix-projection limitation noted in Section 2.5.2.

3.3 Layer 2: Supervised Divergence and Representational Similarity

Out-of-sample R^2 ranged between 0.13 and 0.23, indicating that the supervised models explain only a limited portion of trait variance (Section 4.5). Despite this, primary predictors diverged clearly across traits: agreeableness (Machiavellianism), empathy (psychopathy), and age (narcissism). SHAP interaction analysis found three distinct high-order interaction pairs across the three traits; the single strongest interaction overall belonged to psychopathy (agreeableness $\times$ empathy; strength $= 0.1862$). Symbolic regression discovered three unique functional forms (iteration R^2: Machiavellianism $= 0.093$, psychopathy $= 0.184$, narcissism $= 0.143$). The multi-task network's trait-specific internal representations produced a mean CKA of 0.0369, against a label-permutation null of 0.0912 ($SD = 0.0043$, $p < .0001$) — significantly *lower* than chance alignment, supporting representational distinctiveness.

3.4 Layer 3: Counterfactual Tipping-Point Analysis

Table 6: Primary modifiable feature for crossing the median, by trait

Trait	Primary tipping feature
Machiavellianism	Agreeableness
Narcissism	Agreeableness
Psychopathy	Empathy

An intermediate execution run had mistakenly also flagged agreeableness for psychopathy prior to pipeline debugging; the values above reflect the final reference-run output (Section 4.5 reports this as a reproducibility limitation). Machiavellianism and narcissism converge on agreeableness as their shared behavioral tipping point, while psychopathy depends specifically on empathy — consistent with a possible two-branch structure ("psychopathy" vs. a "Machiavellianism–narcissism cluster"). Counterfactual analysis (Wachter et al., 2017) is not causal inference; these results indicate predictive-model leverage only.

3.5 Layer 4: Semantic Embeddings

Hierarchical clustering on item-text embeddings yielded ARI $= 0.5045$ relative to theoretical labels ($p = .007$ vs. permutation null). Psychopathy and Machiavellianism items each achieved 100% cluster purity; only 75% of narcissism items fell within their dedicated cluster, with the remaining item joining the Machiavellianism cluster. Psychopathy items branched off as an independent primary dendrogram branch, while Machiavellianism and narcissism items intermingled in a shared secondary branch — the "Narcissism Fracture." This is based on only 12 short item texts and a single embedding model (`all-MiniLM-L6-v2`), warranting caution (Section 4.5).

Table 5: Out-of-sample R^2 and primary predictors per trait (XGBoost, representative sample)

Trait	Out-of-sample R^2	Primary predictor	Strongest interaction (strength)
Machiavellianism	0.13–0.23 range	Agreeableness	trait-specific pair
Psychopathy	0.13–0.23 range	Empathy	Agreeableness × Empathy (0.1862)
Narcissism	0.13–0.23 range	Age	trait-specific pair

3.6 Layer 5: Feature Importance and Robustness Testing

The observed SHAP Divergence Index (SDI) was 0.2591, against a 500-iteration pseudo-trait null mean of 0.0953 ($SD = 0.0588$, $p = .0100$), exceeding chance-level partitioning of the same item set. Rashomon Set analysis across XGBoost, Random Forest, and Elastic-Net showed a maximum R^2 variance of $\Delta = 0.019$, indicating architecture-independent predictive performance. GMM clustering of local SHAP vectors showed decreasing BIC with more components (e.g., narcissism: 4097.6 at one component to 3373.0 at five components, with a markedly more gradual decline after two components) — weak evidence for individual-level subtypes. Five-fold cross-validation replicated satisfactorily between community and representative samples (e.g., psychopathy: 0.312 community vs. 0.237 representative); the student sample could not be evaluated due to missing auxiliary variables, so replication claims are restricted to the community–representative comparison.

3.7 Integrated Summary of Evidence

4 Discussion

4.1 Central Finding: Distinctiveness Within a Shared Structure

The five computational layers, together with baseline psychometrics, yield two seemingly contradictory but complementary findings. Every metric examining the three traits in isolation — the DTDD-only network, XGBoost predictor patterns, CKA representations, semantic embeddings, and the SDI — convergently showed distinguishable functional, predictive, and representational patterns. Every metric examining the traits within a broader nomological context — the full 88-item network and latent factor correlations — showed convergence into a shared higher-order structure. This mirrors the prediction of Dark Factor theory (Moshagen et al., 2018): dark traits are surface, partly distinct manifestations of a shared fundamental disposition. The appropriate conclusion is neither "three fully independent traits" nor "one unitary factor," but three empirically distinguishable dimensions nested within a shared higher-order structure — consistent with Vize et al. (2018).

4.2 Machiavellianism and Psychopathy: A Distinction Without a Difference?

Machiavellianism repeatedly occupied an intermediate, bridging position: its highest latent correlation was with psychopathy ($r = 0.62$), and its counterfactual sensitivity to agreeableness was shared with narcissism — consistent with Miller et al. (2017). However, psychopathy's distinctive predictor (empathy, not agreeableness) across Layers 2–3, and its complete semantic separation in Layer 4, indicate that psychopathy retains a distinct functional/predictive identity despite high latent overlap with Machiavellianism, suggesting the overlap may be more a shared-latent-variance phenomenon than a predictive/semantic one.

4.3 The Narcissism Fracture and the Centrality of Agreeableness

The overlap between narcissism and Machiavellianism was independently replicated at both the semantic clustering level (Layer 4) and the counterfactual level (Layer 3), converging on agreeableness. This aligns with Vize et al. (2020), who identified low agreeableness as a leading candidate for the Dark Triad's shared core, contrasted with callousness/low empathy (Paulhus, 2014), here specifically tied to psychopathy. This supports a possible two-branch structure: an "instrumental disagreeableness" pole (Machiavellianism, narcissism) and an "affective callousness" pole (psychopathy) — an interpretation requiring confirmatory testing.

4.4 Position Within the Existing Literature

Using tools entirely independent of CFA, this study converges with the Dark Factor framework (Moshagen et al., 2018) — itself derived from factor-analytic methods in large populations — that dark traits share an underlying disposition. This methodological convergence across unrelated analytic families constitutes a form of convergent validity for the Dark Core hypothesis. Explainable-AI tools (SHAP, CKA) additionally reveal higher-order interactions, nonlinear functional forms, and representational geometry not accessible to linear regression or CFA alone.

4.5 Limitations

1. **Limited explanatory power of supervised models.** Out-of-sample R^2 (0.13–0.23) restricts inferences to nomological overlap rather than the full latent constructs.
2. **Interdependence of the SDI criterion and null.** Both actual scores and pseudo-trait nulls derive from the

Table 7: Direction of evidence across baseline analyses and the five computational layers

Evidence source	Metric	Direction
CFA (three-factor)	CFI $= 0.941$, RMSEA $= 0.077$	Distinctiveness
Latent correlations	$r = 0.61$–0.62	Convergence
Network (12-item)	3 Louvain communities	Distinctiveness
Network (88-item, full)	1 merged community	Convergence (Dark Core)
XGBoost predictors/SHAP	Distinct predictors per trait	Distinctiveness
CKA (neural representations)	0.0369 vs. null 0.0912	Distinctiveness
Semantic embeddings (ARI)	0.5045, $p = .007$	Distinctiveness (narcissism fractured)
Counterfactual tipping points	Shared feature (Mach/Narc)	Partial convergence
SHAP Divergence Index	0.2591 vs. null 0.0953	Distinctiveness
Rashomon robustness	$\Delta R^2 = 0.019$	Stable across architectures

same 12 items, so the SDI null is not fully independent of the measurement system and should not support causal/ontological claims.

3. **Conceptual overlap between predictors and criteria.** Agreeableness and empathy conceptually overlap with the Dark Triad constructs themselves, so observed distinctiveness/overlap may partly reflect content overlap rather than an independent underlying structure.

4. **Small semantic-analysis sample.** Layer 4 rests on only 12 item texts and a single embedding model; stability across models (e.g., MPNet, BGE, E5) remains untested.

5. **Non-causal counterfactual analysis.** Layer 3 identifies predictive-model leverage only, not causal relationships.

6. **Symbolic regression complexity/overfitting risk.** Discovered equations (especially for psychopathy) are complex and treated as supplementary, not primary, evidence.

7. **Student-sample constraints.** Missing auxiliary variables excluded the student sample from predictive modeling and cross-sample replication; replication claims are limited to the community–representative comparison.

8. **Lower psychopathy-subscale reliability.** Lower internal consistency/stability for psychopathy (student-sample $\omega = 0.638$) may partly explain its apparent distinctiveness via measurement error rather than construct differences.

9. **Exploratory, non-preregistered analysis.** Findings are hypothesis-generating and require confirmatory, preregistered replication.

4.6 Suggestions for Future Research

Three directions are proposed: (1) preregistered, confirmatory re-testing of the Narcissism Fracture and agreeableness/empathy centrality in independent samples; (2) robustness testing of Layer 4 across multiple embedding models and, where feasible, the 15-item DTDD variant with additional impulsivity/antisocial items; and (3) integration of the DT^3 framework with longitudinal designs to test temporal and individual-level stability of convergence/divergence patterns.

5 Conclusion

By integrating classical psychometrics with five independent layers of computational evidence — network psychometrics, supervised machine learning, counterfactual analysis, semantic embeddings, and cross-architecture robustness testing — this study provides convergent support for a middle-ground conclusion: Machiavellianism, psychopathy, and narcissism are three empirically distinguishable dimensions nested within a shared higher-order structure, the Dark Core of Personality. This distinctiveness is evident in predictive drivers, neural representations, and (partially) semantic organization, alongside considerable overlap — particularly between Machiavellianism and narcissism around agreeableness. Given the limited explanatory power of the supervised models and the exploratory, non-preregistered nature of this analysis, the findings should be regarded as hypothesis-generating rather than confirmatory. Nonetheless, the convergence of several independent methods toward a single conclusion illustrates the value of integrative psychology–explainable-AI approaches for classic questions of personality structure.

Author Contributions

Sabiheh Khorshidi: psychological conceptualization, interpretation of findings, manuscript writing. Kian Mansouri Jamshidi: design and implementation of the computational pipeline, data analysis, code reproducibility. Both authors contributed to final manuscript writing and approval.

Data and Code Availability Statement

Complete code for preprocessing, baseline analyses, and the five computational layers is available at: `https://github.com/sirkianmj/dark_triad_triangulation`

Conflict of Interest

The authors declare no conflict of interest related to this research.

References

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, *2008*(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). https://doi.org/10.1145/2939672.2939785

Epskamp, S., & Fried, E. I. (2018). A tutorial on regularized partial correlation networks. *Psychological Methods*, *23*(4), 617–634. https://doi.org/10.1037/met0000167

Fisher, A., Rudin, C., & Dominici, F. (2019). All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, *20*(177), 1–81.

Jonason, P. K., & Webster, G. D. (2010). The Dirty Dozen: A concise measure of the Dark Triad. *Psychological Assessment*, *22*(2), 420–432. https://doi.org/10.1037/a0019265

Jones, D. N., & Paulhus, D. L. (2014). Introducing the Short Dark Triad (SD3): A brief measure of dark personality traits. *Assessment*, *21*(1), 28–41. https://doi.org/10.1177/1073191113514105

Knitter, L. A., Hoffmann, J., Eid, M., & Koch, T. (2025). Measuring the dark triad: A meta-analytical SEM study of two prominent short scales. *Frontiers in Psychology*, *15*, 1469970. https://doi.org/10.3389/fpsyg.2024.1469970

Kornblith, S., Norouzi, M., Lee, H., & Hinton, G. (2019). Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning* (pp. 3519–3529). PMLR.

Leberecht, M., Nedderhoff, A., Zitzmann, S., & Hecht, M. (2026). Comparing machine learning methods for predicting dark triad personality traits using social media text data. *Journal of Research in Personality*, *120*, Article 104690.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30* (pp. 4765–4774).

Miller, J. D., Hyatt, C. S., Maples-Keller, J. L., Carter, N. T., & Lynam, D. R. (2017). Psychopathy and Machiavellianism: A distinction without a difference? *Journal of Personality*, *85*(4), 439–453. https://doi.org/10.1111/jopy.12251

Moshagen, M., Hilbig, B. E., & Zettler, I. (2018). The dark core of personality. *Psychological Review*, *125*(5), 656–688. https://doi.org/10.1037/rev0000111

Muris, P., Merckelbach, H., Otgaar, H., & Meijer, E. (2017). The malevolent side of human nature: A meta-analysis and critical review of the literature on the Dark Triad (narcissism, Machiavellianism, and psychopathy). *Perspectives on Psychological Science*, *12*(2), 183–204. https://doi.org/10.1177/1745691616666070

O'Boyle, E. H., Forsyth, D. R., Banks, G. C., Story, P. A., & White, C. D. (2015). A meta-analytic test of redundancy and relative importance of the Dark Triad and five-factor model of personality. *Journal of Personality*, *83*(6), 644–664. https://doi.org/10.1111/jopy.12126

Papageorgiou, K. A., Benini, E., Bilello, D., Gianniou, F.-M., Clough, P. J., & Costantini, G. (2019). Bridging the gap: A network approach to Dark Triad, Mental Toughness, the Big Five, and perceived stress. *Journal of Personality*, *87*(6), 1250–1263. https://doi.org/10.1111/jopy.12472

Paulhus, D. L. (2014). Toward a taxonomy of dark personalities. *Current Directions in Psychological Science*, *23*(6), 421–426. https://doi.org/10.1177/0963721414547737

Paulhus, D. L., & Williams, K. M. (2002). The Dark Triad of personality: Narcissism, Machiavellianism, and psychopathy. *Journal of Research in Personality*, *36*(6), 556–563. https://doi.org/10.1016/S0092-6566(02)00505-6

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing* (pp. 3982–3992). https://doi.org/10.18653/v1/D19-1410

Shukla, M., & Upadhyay, N. (2025). Cold hearts and dark minds: A systematic review and meta-analysis of empathy across dark triad personalities. *Frontiers in Psychiatry*, *16*, 1546917. https://doi.org/10.3389/fpsyt.2025.1546917

Vize, C. E., Collison, K. L., Miller, J. D., & Lynam, D. R. (2020). The "core" of the dark triad: A test of competing hypotheses. *Personality Disorders: Theory, Research, and Treatment*, *11*(2), 91–99. https://doi.org/10.1037/per0000386

Vize, C. E., Lynam, D. R., Collison, K. L., & Miller, J. D. (2018). Differences among Dark Triad components: A meta-analytic investigation. *Personality Disorders: Theory, Research, and Treatment*, *9*(2), 101–111. https://doi.org/10.1037/per0000222

Wachter, S., Mittelstadt, B., & Russell, C. (2017). Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law & Technology*, *31*(2), 841–887.

Wehner, C., Ziegler, M., Gödeke, W., & Lämmle, L. (2021). Further inflaming the discussion or cooling down feelings? A network analysis of the Dark Triad and the Five Factor Model of personality. *Personality and Individual Differences*, *175*, Article 110717. https://doi.org/10.1016/j.paid.2021.110717

Živná, K., Novak, L., Zidkova, R., Helvich, J., Tavel, P., & Malinakova, K. (2025). Impulsivity, psychopathy, and the dark triad: A Czech validation of the

Dirty Dozen scale. *BMC Psychology*, *13*, Article 1170. https://doi.org/10.1186/s40359-025-03371-y